

## Problem-Solution Fit

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 29 June 2026                               |
| <b>TeamID</b>        | xxxxxx                                     |
| <b>ProjectName</b>   | Voice Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 5 Marks                                    |

### Problem-SolutionFit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

|                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>1. CUSTOMER SEGMENT(S)[CS]</b></p> <ol style="list-style-type: none"> <li>1) Students preparing for examinations and academic assessments</li> <li>2) Job seekers preparing for technical interviews and aptitude rounds</li> <li>3) Educators and Trainers evaluating students' conceptual understanding</li> <li>4) Educational Institutions and Training Organizations</li> </ol> | <p><b>6. CUSTOMER LIMITATIONS</b><br/>EG.BUDGET,DEVICES [CL]</p> <ol style="list-style-type: none"> <li>1) <input type="checkbox"/> Lack of immediate and objective feedback on spoken explanations</li> <li>2) <input type="checkbox"/> Difficulty assessing conceptual understanding independently</li> <li>3) <input type="checkbox"/> Nervousness and low confidence during interviews or presentations</li> <li>4) <input type="checkbox"/> Frequent use of filler words and communication issues</li> <li>5) <input type="checkbox"/> Limited access to mentors for regular evaluation</li> <li>6) <input type="checkbox"/> Need for continuous practice without personalized assessment</li> </ol> | <p><b>5. AVAILABLE SOLUTIONS</b><br/>PROS &amp; CONS [AS]</p> <ol style="list-style-type: none"> <li>1) <b>Traditional Self-Practice</b></li> <li>2) <b>Pros:</b></li> <li>3) Free and easy to access</li> <li>4) Flexible learning pace</li> <li>5) <b>Cons:</b></li> <li>6) No objective evaluation</li> <li>7) Difficult to identify conceptual mistakes</li> <li>8) <b>Teacher / Mentor Feedback</b></li> <li>9) <b>Pros:</b></li> <li>10) Personalized guidance</li> <li>11) Expert evaluation</li> <li>12) <b>Cons:</b></li> <li>13) Time-consuming</li> <li>14) Limited availability</li> </ol> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- 15) May not provide instant feedback
- 16) **Online Learning Platforms**
- 17) **Pros:**
- 18) Large collection of learning resources
- 19) Convenient and accessible
- 20) **Cons:**
- 21) Focus mainly on content delivery
- 22) Limited evaluation of spoken conceptual understanding
- 23)

## 2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- 1) ☐ Difficulty determining whether a spoken explanation is conceptually correct **(High)**
- 2) ☐ Lack of immediate and objective feedback on conceptual understanding **(High)**
- 3) ☐ Poor communication due to filler words, hesitation, and low confidence **(Medium)**
- 4) ☐ Limited opportunities to practice with personalized evaluation before interviews or

## 9.PROBLEMROOT/ CAUSE [RC]

- 1) Lack of personalized evaluation tools
- 2) No immediate feedback on spoken explanations
- 3) Difficulty identifying communication weaknesses
- 4) Limited opportunities for guided practice

## 7. BEHAVIOR + ITS INTENSITY [BE]

- ☐ Practices concepts before exams or interviews
- ☐ Watches educational videos to improve understanding
- ☐ Rehearses presentations multiple times
- ☐ **Intensity:** High (especially during exams, placements, and presentations)

presentations (**High**)

### **3. TRIGGER TO ACT [TR]**

- 1) ☐ Upcoming examinations or viva assessments
- 2) ☐ Technical interviews or placement preparation
- 3) ☐ Classroom presentations or seminars
- 4) ☐ Desire to improve communication and conceptual understanding

### **4. EMOTIONS BEFORE / AFTER [EM]**

- 1) **Before**
- 2) Confused
- 3) Nervous
- 4) Unsure about understanding
- 5) Lacking confidence
- 6) **After**
- 7) Confident
- 8) Motivated
- 9) Better prepared
- 10) Aware of strengths and areas for improvement
- 11)

### **10. YOUR SOLUTION [SL]**

**Voice-Based Concept Understanding Analyser (VBCUA) is an AI-powered educational assessment platform that evaluates users' spoken explanations using OpenAI Whisper for speech transcription, Sentence-BERT for semantic similarity analysis, and Librosa for speech fluency analysis. The system measures conceptual understanding, identifies communication weaknesses, provides an understanding score with qualitative feedback, visualizes speech metrics, and generates downloadable PDF reports to help learners improve both knowledge and presentation skills.**

### **8. CHANNELS OF BEHAVIOR [CH]**

- 1) **ONLINE**
- 2) Educational platforms (Coursera, Udemy, NPTEL)
- 3) YouTube
- 4) ChatGPT and AI learning tools
- 5) College LMS / Google Classroom
- 6) **OFFLINE**
- 7) Classroom lectures
- 8) Study groups
- 9) College laboratories
- 10) Faculty mentoring sessions
- 11)
- 1)